

Masking a Field's Loss Does Not Isolate That Field

Two rejected LoRA runs, and what an unsupervised output field does when its neighbor is tuned

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Preprint. This is the named version of a paper prepared for peer review. The anonymized submission build is identical in content; only the author block, the companion-paper citation keys and the anonymization note differ.

Draft 2 adds §5: a three-arm target-module ablation that discharges the principal limitation of Draft 1. One of the three isolation routes §6 offered is closed by it.

Written in third person for double-blind. Model, corpus and infrastructure names are given; no author, institution or repository is identified.

Abstract

Structured-output guard models emit several fields per inference, typically a severity verdict and a category. When one underperforms, a natural experiment is to mask its loss and tune the other, reasoning that withholding supervision preserves behavior.

It does not. We report two rejected fine-tuning runs on a 0.6B guard model. Run 1 supervised severity with the binary targets its corpus supports and extinguished its third severity class, 49 Controversial verdicts falling to 0, converting a documented operator lever into a no-op that still reported success. Run 2 masked the severity line in the loss while keeping it in the input, and scored 17 points of overall recall below run 1 and 23 adversarial points below the untuned base model. The masked field did not hold still: Controversial overproduced at 187 against a base of 49, Unsafe collapsed from 650 to 367, and the model began emitting its own category vocabulary in the severity slot.

Neither mechanism is new. Concurrent work reports a masked field degrading under the same configuration, and binary collapse is standing practice in guardrail dataset design. Three narrower findings survive: the per-class distribution of a masked categorical enum against base-model counts; the direction of the effect, opposite in sign to the concurrent result whose unmasked baseline supervises a degenerate target; and the governance consequence, that a masked field is a policy surface whose silent movement can disable an audited control.

Three further runs then test the repair the first two implied. Holding corpus, base model, rank and schedule fixed and varying **only** which projections the adapter touches, we train an attention-only arm, a disjoint MLP-only arm, and a full control. **All three extinguish the class**, on a second evaluation set where the untuned model emits 122 Controversial verdicts rather than 49. The manipulation is not inert — it swings the false-positive rate by nearly a factor of three, from 0.059 to 0.165 — it simply does not touch this. Choosing better target modules is therefore not available as a repair, and of the three isolation routes we identified, only a genuinely separate adapter and a corpus carrying a third label remain.

We report all five runs, the automated gate's verdicts, and the corpus property that blocked the intended repair.

1. Why report two runs that were rejected

Guard models are increasingly the primary runtime control in agentic deployments, and they are increasingly structured-output models rather than binary classifiers: a severity verdict *and* a category, parsed downstream by policy that may treat either as dispositive.

That structure invites a specific experiment. When one field is weak, tune the other and leave the first alone. The intuition is that supervision is what changes a model, so withholding supervision should preserve behavior.

We ran that experiment. It failed, and it failed in a direction that is worth more than a promotion would have been. Both runs were rejected by an automated parity gate before any promotion decision reached a human, and the gate's two verdicts distinguish precisely between *no improvement demonstrated* and *this made it worse* — a distinction the literature on fine-tuning guards does not usually get to make, because unsuccessful runs are rarely reported at all.

Contributions.

△ **Novelty, bounded first.** That loss masking fails to isolate a field under a shared adapter is **documented concurrently**, and that binary targets produce binary behavior is **established practice**. §8 withdraws the broader claim an earlier draft made and cites the work that refutes it. Four narrower contributions survive:

1. **The distributional signature.** The per-class distribution of a masked *categorical enum* measured against the base model's own counts — overproduction of one class, collapse of another, and migration of the neighboring field's vocabulary into the masked slot (§4.2).
2. **The direction.** Masking cost 17 recall points *against supervising the same field with wrong targets*. That is the opposite sign to the concurrent result, whose unmasked baseline supervises a degenerate target, and the disagreement is informative rather than contradictory (§4.1, §8).
3. **The governance consequence.** A masked field is a policy surface. Its silent movement converted a documented operator control into a no-op that reported success (§3.3). We are aware of no treatment of masked-field drift as an auditability failure rather than a quality failure.
4. **The corpus constraint and the prescription.** The signal needed to supervise three-valued severity exists only where it cannot be trained on (§3.5), and isolation therefore requires separate parameters rather than a masking flag (§6).
5. **The target-module ablation.** Two disjoint families of projections, each independently sufficient to destroy the class, on a configuration whose collapse reproduces to within one item across hardware and ten days (§5). This converts the paper's central mechanistic claim into an empirical one and closes the cheaper of the two structural repairs — the reading of "separate parameters" that means *choose different target modules*, as opposed to *use a second adapter*.

2. Setup

Model. A 0.6B parameter generative guard model, fine-tuned with LoRA. A same-size untuned baseline is retained deliberately: without a same-size comparator, a rejection cannot distinguish *the fine-tune did not work* from *this model is smaller than the one it is being compared to* — different findings with different next actions.

Output format. Each inference emits a severity line and a category line. Severity is three-valued in the base model's vocabulary — **Safe**, **Unsafe**, **Controversial**. Downstream policy treats a gating category as harmful in its own right, and separately allows an operator to decide whether **Controversial** counts as harmful.

Corpus. A public guard training mixture of 38,694 rows. **Its severity targets are binary:** the rendering function emits only **Safety: Unsafe** or **Safety: Safe**, and the corpus contains 19,349 of one and 19,345 of the other. There is no third value anywhere in the training targets.

Evaluation. The full held-out test split, 1,699 items, scored on recall first, with per-category recall and the confusion matrix reported alongside.

△ **Reported precision of every figure in this paper.** Headline rates are reported as recorded, to four decimal places. **Per-class counts were reconstructed from the recorded rates and denominators** and are accurate to approximately one row; they are reported as integers because that is how they were reconstructed, not because they were counted. Differences of one or two in a class count carry no weight in any argument here. The two that do carry weight — 49 **Controversial** verdicts falling to 0, and 187 against a base of 49 — are order-of-magnitude effects that survive the uncertainty by a wide margin. §7 states why the underlying per-item outputs cannot be re-derived. An automated gate compares a candidate against a measured baseline on the same lane and precision, and rejects on either an unimproved headline or a per-category floor breach.

Compute. Run 1: a single A100, 38,694 rows, one epoch, 2 hours 28 minutes, final training loss 0.0101. The runs are small; the findings are about direction, not scale.

3. Run 1 — supervising a three-valued field with binary targets

3.1 The gate rejected it, and was right twice

VERDICT: REJECT (exit 1)

- recall 0.8329 against 0.8554 is within sampling noise at these counts (this set could resolve about 0.036). No improvement was demonstrated, which is not the same finding as a regression.
- per-category recall fell below a measured baseline floor: others 0.7653 below 0.7857; social stereotypes and unfair discrimination 0.6184 below 0.7237

The first clause is a discipline worth naming: **the gate refuses to call an unresolvable difference an improvement**, and says so in the language of what the test set can resolve.

3.2 Against the same-size baseline, the fine-tune made the model worse

	untuned 0.6B	tuned	Δ
Overall recall	0.8488	0.8329	−0.0159
False-positive rate	0.0624	0.0519	−0.0105
Precision	0.9156	0.9276	+0.0120
Controversial verdicts emitted	49	0	extinguished

untuned 0.6B	tuned	Δ
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Recall down, false positives down, precision up: **the adapter learned to be more permissive.** For a deny gate that is the wrong direction, and it is the direction that fails silently — a guard that blocks less looks better on every metric an operator watches casually.

The mechanism is the target vocabulary. One epoch of binary targets extinguished the third severity class entirely. The rendering function's own reasoning had been that inventing a **Controversial** target the labels do not distinguish would teach a distinction the data never made, and that downstream policy scoring **Controversial** as harmful "would quietly absorb the difference." **It did not absorb it.** Every prompt that would have been **Controversial** — and therefore caught, because policy scores it harmful — is now **Safe** and missed.

3.3 The second-order failure: a documented lever became a no-op

This matters more to a governance product than the recall delta.

The deployment exposes an operator policy row: *does a borderline verdict count as harmful?* On the untuned model that lever moves recall from **0.8488 to 0.8011** — a real, documented control with a measurable effect.

On the tuned model, the same lever reports **0.8329, identical to the headline row**, because there are no **Controversial** verdicts left for it to act on.

A documented operator control silently became inert, and nothing in the system announced it. No error, no warning, no change in any interface. The lever is still in the configuration, still in the documentation, and now does nothing. For a system whose purpose is to be auditable, a control that reports success while having no effect is a worse failure than one that breaks loudly.

3.4 Targeting the weak category did not fix the weak category

The run's stated purpose was to improve three underperforming classes. Two moved marginally; the largest collapsed.

Class (all three targeted)	Untuned	Tuned	Δ
fraud_assisting_illegal_activities	0.7833	0.8000	+0.0167
others	0.7551	0.7653	+0.0102
social_stereotypes_and_unfair_discrimination	0.7763	0.6184	-0.1579

The finding is not that LoRA does not work here. It is that **a corpus whose targets are binary trains a binary model**, and weak-category selection does not survive contact with a three-valued output space.

3.5 The repair that was not available

Two repairs were on the table. Render **Controversial** for rows the corpora mark borderline, or stop supervising severity at all.

The first is not implementable from either corpus. The training mixture's borderline signal — an agreement column distinguishing unanimous from split annotator judgments — **exists only in the test split.** The train split does not carry it, and the second corpus in the mixture has no equivalent column at all.

That is a corpus-level constraint with a general shape: **the signal needed to supervise a three-valued severity output exists where it cannot be trained on and is absent where it could be**. Any practitioner attempting a graded severity output from these corpora will hit the same wall.

Correction, 2026-08-30: the constraint is corpus-specific, not general. The paragraph above is accurate about the two corpora in this program's mixture, and an earlier draft over-generalized it to "any practitioner attempting a graded severity output from these corpora." A survey of public alternatives refutes that. The Aegis AI Content Safety Dataset 1.0 carries an explicit third severity value, **Needs Caution**, in its **training** split -- 1,658 occurrences in a 4,000-row sample -- alongside five per-annotator label columns that supply exactly the annotator-agreement signal this section reports as available only in WildGuardMix's test split.

So the accurate statement is narrower and more useful. **Supervising a three-valued severity output is possible; it is not possible from the mixture used here.** The corpus-selection decision that produced the failure is the finding, not a property of the field. A practitioner who needs graded severity should choose a corpus that labels it, and one exists.

This correction does not rescue the runs. Both were trained on the mixture described above and both failed for the reasons given. It removes an over-claimed generalization from the analysis of why.

So supervising severity at all means supervising it binary. Which left the second option.

4. Run 2 — masking the field, and what the field did

The reasoning was explicit and, we thought, conservative: mask the severity line in the loss, keep it in the input, and the base model's severity distribution should be left intact while the category line is tuned. Downstream policy treats a gating category as harmful independently, so catch rate could improve through the category path alone.

4.1 The measurement

Run	Overall recall	Adversarial recall	Severity verdicts emitted
Untuned 0.6B	0.8488	0.7918	controversial 49, safe 1000, unsafe 650
Run 1 — severity supervised	0.8329	0.7947	controversial 0 , safe 1022, unsafe 677
Run 2 — severity masked	0.6804	0.5572	controversial 187 , safe 1132, unsafe 367 , <i>plus category words</i>

Masking was 17 points of overall recall worse than supervising the field binary, and 23 adversarial points below the untuned model. The gate rejected run 1 as *within sampling noise* and run 2 as *recall regressed beyond sampling noise* — correctly separating "no improvement shown" from "this made it worse."

4.2 The distributional signature

The recall number says the run failed. The verdict distribution says **why**, and it is the paper's central evidence.

An isolated field would hold its distribution. Severity did not hold anything:

- **Controversial returned but overproduced** — 187 against the base model's 49, nearly four times.
- **Unsafe collapsed** — 367 against 650.
- **The model began emitting its own category vocabulary in the severity slot** — strings such as `non-violent illegal acts`, `others`, `violent` appearing where `Safe`, `Unsafe` or `Controversial` belong.

That third observation is the one that settles it. A field that was merely *uncorrected* would drift within its own vocabulary. A field filling with **the neighboring field's tokens** has not been left alone; it has been actively reshaped by training it was supposedly excluded from.

4.3 The mechanism

A LoRA adapter modifies the query, key, value, output, gate, up and down projections across every layer. **Both output fields are produced by those same weights.**

Not computing a loss on severity did not hold severity still. It only stopped *correcting* severity while training moved it anyway. Masking removes a gradient signal; it does not build a wall — and in the absence of a corrective signal, the field drifted toward the token distribution that training was actively reinforcing next door.

Stated generally: **under a shared low-rank update there is no "rest of the model" to hold constant.** Every output field rides the same weights. Loss masking is not a form of parameter isolation and should not be reasoned about as one.

5. Runs 3–5 — retargeting the adapter, and the limitation that closes

§7 of an earlier draft of this paper named a limitation and made a prediction under it:

One adapter configuration tested. A different rank or a different target-module set might reduce the leakage. **It should not eliminate it**, since the projections producing both fields overlap under every standard configuration we are aware of — but we did not test that, and the claim is therefore mechanistic rather than empirical.

We have now tested it. The prediction holds, and the route it was hedging is closed.

5.1 Design

Three arms, identical in everything but the adapter's target modules. Same base model, same 11,272-row training corpus, rank 16, alpha 32, bf16, one epoch, single A100.

arm	target modules	rationale
A control	<code>q,k,v,o,gate,up,down</code>	the configuration of run 1, retrained
B attention	<code>q,k,v,o</code>	attention projections only
C MLP	<code>gate,up,down</code>	feed-forward projections only

B and C are **disjoint**. If the severity field is produced by one family and not the other, exactly one arm should preserve it.

A **stopping rule was fixed before arm A was dispatched**: if the control did not reproduce the severity collapse, the diagnosis was wrong and arms B and C would be cancelled rather than interpreted. We record it because it is the condition under which we would have reported nothing.

A different evaluation set from runs 1 and 2. Runs 1 and 2 were scored on WildGuardTest, where the untuned 0.6B emits 49 **Controversial** verdicts. Runs 3–5 are scored on ExpGuardTest, 2,275 items, where it emits **122**. The collapse therefore reproduces on a second evaluation set with a 2.5× larger severity population, which is a stronger statement than a repeat on the original set.

5.2 The control reproduces to within one item

Arm A is a retrain of run 1's configuration, ten days later, on different hardware.

	baseline-only	candidate-only	unsafe	controversial
original run	31	115	1038	122 → 0
arm A (retrain)	31	116	1038	122 → 0

Identical **unsafe** count, identical baseline-only count, candidate-only differing by one. The **safe** counts differ by exactly seven — which is exactly the number of items whose backend call failed in the second run. Outside those seven the two runs agree item for item.

The collapse is therefore a property of the configuration, not of the run. Not a seed, not an unlucky checkpoint. This is what makes the ablation interpretable at all: there is no run-to-run variance for a module effect to hide behind.

5.3 Every arm loses the class

arm	modules	recall	FPR	safe	unsafe	controversial	base-only	cand-only	exact p
baseline	—	0.7150	0.0854	1290	863	122	—	—	—
A control	all seven	0.7842	0.0589	1230	1038	0	31	116	<0.0001
B attention	q,k,v,o	0.8439	0.1649	1047	1227	0	2	164	<0.0001
C MLP	gate,up,down	0.7930	0.0864	1191	1084	0	19	117	<0.0001

Paired exact-binomial McNemar over shared item ids, positives only; items either run failed to classify are excluded rather than scored as misses. Arm A's recall is inflated by roughly the seven backend failures, which the harness scores fail-closed as harmful; the paired counts exclude them.

Attention-only destroys the class. MLP-only destroys the class. Two disjoint module families, each independently sufficient. There is no third family to try.

5.4 The manipulation worked — on something else

A null result is only informative if the intervention did anything at all. It did:

- **Attention-only is badly miscalibrated.** FPR nearly doubles against the baseline, 0.0854 → **0.1649**, and the arm calls 1,227 of 2,275 items unsafe. Its 0.8439 recall is the highest figure in this program and is substantially bought with false positives.
- **MLP-only is FPR-neutral** — 0.0864 against 0.0854 — while gaining recall.
- **All seven improves FPR**, to 0.0589.

Module choice swings the false-positive rate by nearly a factor of three and leaves the severity class at zero in every arm. The intervention is potent; it is simply not potent on this.

5.5 What this closes, and what it does not

§6 offered three routes to isolation. This experiment resolves one of them and touches neither other:

1. **A second adapter carrying the severity field — untested, and now the only surviving structural route.** Note that it is *not* what this experiment tested: runs 3–5 varied which modules a *single* adapter touches, never how many adapters exist. The distinction matters and was blurred in our own planning before the runs.
2. **Target modules that do not carry severity — closed.** No such module set exists among the projections a LoRA adapter can address here.
3. **A corpus that can supervise all three values — untouched by this experiment,** and per §3.5 the Aegis dataset does carry a third value in its train split. That route remains open and is the cheaper of the two survivors, because it requires no architectural change.

The mechanistic claim of §7 is now an empirical one, and the weaker reading of "separate parameters" — choose better target modules — is refuted. Only the stronger reading survives.

6. What follows

The default was restored to supervising severity binary — the less bad of two measured losses, and it at least keeps the severity line well-formed. **That is not a fix**, and we record it as a retreat rather than a resolution.

Isolation requires separate parameters, not a flag. Three routes remain, and none of them is a third setting of the masking option:

1. **A second adapter** carrying the severity field, with its own parameters.
2. **Target modules that do not carry severity** — which requires knowing which projections produce which field, and is a real open question at this scale.
3. **A corpus that can supervise all three severity values** — which, per §3.5, does not currently exist in the public guard mixtures.

A note on gates. Both runs were stopped by an automated parity gate before a human decided anything. The gate's value here was not that it caught a bad model; it is that it produced two *differently worded* rejections, and the difference between them — unresolvable versus regressed — is what made the second run interpretable as evidence rather than as noise. A promotion gate that emits a single boolean would have discarded the finding.

7. Limitations

Single model, single scale. One 0.6B model, one adapter configuration. We do not claim the effect size generalizes; we claim the mechanism does, because it follows from weight sharing rather than from any property of this model.

One adapter configuration tested — DISCHARGED for target modules, standing for rank. This limitation previously read: *"A different rank or a different target-module set might reduce the leakage. It should not eliminate it... but we did not test that, and the claim is therefore mechanistic rather than empirical."* §5 tests the target-module half and the prediction holds: attention-only and MLP-only each eliminate nothing and each destroy the class outright, so for target modules the claim is now empirical. **The rank half is still untested.** All three arms ran at rank 16, alpha 32; we do not know what a much larger or much smaller adapter does, and §5's result gives no reason to expect rank to behave like target-module choice.

The evaluations were re-run, and they reproduce. An earlier version of this section reported that the raw per-item outputs no longer existed: the results directory had been excluded from version control and never committed, so every figure survived only as recorded prose and as hand-transcribed literals whose per-class counts were back-computed from published rates.

Both baseline evaluations were re-executed on 2026-08-30 against the same pinned configuration -- num_ctx 8192, seed 0, temperature 0, fail-closed, the full 1,699-item held-out split, both models at Q4_K_M with the digest recorded in the original run. The regenerated figures agree with the recorded ones on every reported metric:

	recorded	re-run	delta
4B overall recall	0.8554	0.85676	+0.0014
4B precision	0.9241	0.92418	+0.0001
4B adversarial=false recall	0.8886	0.88862	+0.0000
4B adversarial=true recall	0.8152	0.81818	+0.0030
0.6B overall recall	0.8488	0.84615	-0.0026
0.6B Controversial -scored-safe recall	0.8011	0.80106	-0.0000

The two per-category floors cited in the original gate rejection -- **others** at 0.7857 and **social_stereotypes_and_unfair_discrimination** at 0.7237 -- reproduce **exactly**. Severity verdict counts reproduce to within one or two rows (**Controversial** 48 against a recorded 49; safe 1,002 against 1,000; unsafe 649 against 650), which is precisely the one-row precision section 2 claimed for the reconstructed figures and had no way to verify at the time.

Consequence. The per-item records exist again: 1,699 rows per model, each carrying the sample identifier, expected and predicted labels, emitted severity, gated categories and error state. This paper ships an artifact rather than an apology, and its baseline numbers are now measured twice, four months apart, on the same configuration.

What this does not restore. The re-run regenerates the two *baseline* evaluations. The adapters from the two rejected training runs were not retained, so the tuned-model figures in sections 3 and 4 remain as recorded and are **not** independently re-verified. Reproducing those requires re-training, which is out of scope here and is stated as an open item rather than absorbed.

Corpus access. Both corpora are access-gated, and one carries a commercial-use restriction whose license and gate form disagree. Reproduction requires clearing that.

8. Related work

An earlier draft of this section claimed that no treatment existed of what happens to an unsupervised output field when a neighboring field is tuned under a shared adapter. **That claim does not survive a literature check and has been withdrawn.** *Reasoning-Trace Collapse* (arXiv:2605.21127) runs the same configuration — a rank-16 LoRA over the standard attention and MLP projections, four open-weight reasoning models, two-field completions of the form (reasoning, answer), the reasoning field excluded from the cross-entropy but left in the target while the answer field is supervised — and it measures the masked field, which does not hold still: valid-trace rates fall to 61–67 percent and the residual failure mode shifts from empty traces to truncated ones; that work calls masking a partial mitigation. Three narrower components survive, and the contribution below is stated at that width.

Run 1 is the better documented of the two, and is claimed only as a confirming instance. That

binary supervision produces binary behavior is standing practice rather than a discovery: the Aegis guardrail datasets carry an explicit third *Needs Caution* label and ship two LoRA-tuned LlamaGuard variants differing only in whether that label was folded into Safe or into Unsafe (Ghosh et al., NAACL 2025). Minority collapse supplies the mechanism — above an imbalance threshold the classifiers for minority classes collapse onto a mutually indistinguishable direction, and a class with no training targets at all is the degenerate case (Fang et al., PNAS 2021). The result was also predicted in print: Choudhary et al. (COLM 2026) merge a four-level graded harm classifier with a binary refusal model, find graded classification collapsing while refusal transfers under every standard merge rule, and close by naming risk triage and content classification as capabilities vulnerable to being overwritten by a coarser objective. Run 1 confirms that prediction in the more common single-run setting. What the reviewed work does not report is the deployment form of §3.3: an output schema and a documented operator lever that both remain in place after the class they act on has been eliminated.

The mechanism behind run 2 is likewise established, and existing vocabulary is adopted rather than any coined. Multi-task learning names cross-objective degradation through shared parameters *negative transfer*, and its anti-correlated signature the *seesaw phenomenon*, with separation of shared from task-specific parameters as the standard remedy (Tang et al., RecSys 2020). Continual learning gives the sharper form: under a moving backbone, classes receiving no loss term still have their feature distributions drift, which is why *semantic drift* compensation exists (Yu et al., CVPR 2020; Goswami et al., ECCV 2024). The same holds for LoRA specifically — adaptation degrades held-out capability that no loss protects, and the working remedy is an explicit preservation term over the non-target distribution rather than its omission (Xu et al., arXiv:2605.29498) — while interference under a single shared adapter is documented with uniformly parameter-space remedies (arXiv:2504.07448; arXiv:2601.09684). *Parameter isolation* is itself a term of art in continual learning, so the formulation in §4.3 assembles two existing terms and names nothing new. Closest of all, multi-task partially supervised learning makes zeroing an unlabeled task's loss its standard recipe, and its pseudo-supervision machinery exists because that recipe does not suffice (Kokkinos, CVPR 2017; Li et al., CVPR 2022).

What none of this supplies is the inverse measurement. Masking is evaluated there by its effect on the supervised objective (arXiv:2404.07965; Huerta-Enochian and Ko, EMNLP 2024), or at the granularity of a task, a held-out capability, or a broad disposition (arXiv:2502.17424). No located work reports what the masked positions themselves go on to emit, as a per-class distribution measured against the base model's. Three claims are therefore advanced: that per-field distributional measurement (**Controversial** 49 to 187, **Unsafe** 650 to 367); the direction of the result, in which masking cost 17 points of recall against supervising the field with targets known to be wrong — the opposite sign to arXiv:2605.21127, whose unmasked baseline supervises a degenerate empty target that masking removes; and the governance consequence, that a masked field is a policy surface whose silent movement invalidates a documented control. The vocabulary migration in §4.2 already has a name — a *placement error*, a correct value emitted at the wrong position, in the taxonomy of Zhang et al. (arXiv:2608.25358) — and that name is adopted here.

The absence asserted above is bounded by how it was tested: three independent English-language sweeps of arXiv, alphaXiv and the open web, covering the PEFT and loss-masking literature, the multi-task interference and partial-supervision literature, and guard-model fine-tuning, with full texts read for the closest hits. No systematic ACL Anthology or citation-graph sweep was run; a forward sweep on arXiv:2605.21127 or the partially-supervised multi-task line could plausibly close the remaining gap, and the claims above are scoped to survive that.

9. Conclusion

Masking a field's loss does not isolate that field under LoRA. The masked field came unmoored, overproduced one class, collapsed another, and filled with its neighbor's vocabulary — costing 17 points of recall against the alternative it was meant to improve on.

The prior run that motivated the masking is the more useful finding for anyone shipping a guard: a corpus with binary targets trains a binary model, and it will **extinguish a third output class without announcing it** — silently converting a documented operator control into a no-op that reports success.

We publish both because they cost two runs and a week, the explanations are each one sentence, and we could not find either sentence written down.

Production notes (strip before submission)

Status: complete, no placeholders except the related-work citations. Every number is from the recorded run documents. This is the only paper in the program whose results section did not require new compute — because the runs already happened and were rejected.

Before submission.

1. Complete §8 from full reads. The claim of novelty is stated narrowly and must be checked against the PEFT literature specifically.
2. ~~Confirm the back-computed per-slice counts against a surviving log~~ — **closed by the second route.** The logs are gone and cannot be recovered; §2 now states the reported precision explicitly and identifies which two figures the argument actually rests on. No table in this paper asserts a count it did not reconstruct.
3. Decide the venue. Two deadlines were verified against the calls for papers on 2026-08-30: **AIWILD @ NeurIPS 2026** (Third Workshop on Agents in the Wild), submissions **5 September 2026 AoE**, non-archival, 9 pages regular or 4 pages short — the immediate target, and the workshop whose ICML 2026 edition published a comparable three-negative-results safety paper; and **IEEE SaTML 2027**, abstract **22 September 2026**, paper **29 September 2026**, 12 pages of body text, with an anonymized artifact due within three days of submission. SaTML's position/short track fits, and it shares the 29 September deadline with T-12 — which is either efficient or overcommitted depending on the verification outcome on that paper. TMLR is the archival fallback: rolling submission, significance explicitly not a bar, and explicit rejection of papers that incorrectly claim novelty over existing published work.

What makes this publishable where the companion empirical paper is not. That paper's findings need runs that have not happened. This paper's finding *is* the runs that happened and failed. The evidence is complete, the citations are done, and **nothing in it waits on a person or a GPU.**

Status: submittable. No placeholders, no unfilled figures, no external dependency. The nearest deadline is **AIWILD @ NeurIPS 2026 on 5 September** — six days — at 4 pages short or 9 regular, non-archival, which makes it a low-cost first outing that does not spend the archival version. SaTML follows on 29 September, and TMLR is the archival fallback whose stated policy — significance not a bar, and explicit rejection of papers overclaiming novelty — suits a narrowed negative result unusually well.